

System Topology - DocuMan Application

Architecture Diagram

```

graph TB
    subgraph "External Access"
        User[👤 User]
        Laptop/Mobile
        Any Device
        Internet((Internet))
    end

    subgraph "Cloudflare Network"
        CF[🌐 Cloudflare Tunnel]
        Quick Tunnel
        HTTPS
        Edge[Cloudflare Edge]
        Location: dps01/sin15
    end

    subgraph "Local Machine – MacBook Pro"
        subgraph "Port Forwarding Layer"
            Socat[🔄 Socat]
            Port 5173
            TCP Forwarder
        end

        subgraph "Web Server Layer"
            Nginx[🌀 Nginx]
            Port 3000
            Reverse Proxy
        end

        subgraph "Application Layer"
            Frontend[🌈 Frontend]
            React SPA
            Static Files
            Backend[🔗 Backend]
            Node.js Express
            Port 5001
        end

        subgraph "Data Layer"
            DB[(🗄️ PostgreSQL)]
            Database
            documan_db
        end

        subgraph "File Storage"
            Uploads[📁 File Storage]
        end
    end

```

```

uploads/]
    end
end

%% External connections
User -->|HTTPS Request| Internet
Internet -->|Encrypted Tunnel| Edge
Edge -->|HTTP/2 Protocol| CF

%% Cloudflare to local
CF -->|http://localhost:5173| Socat

%% Port forwarding
Socat -->|Forward to Port 3000| Nginx

%% Nginx routing
Nginx -->|"Route: /"
Serve Static| Frontend
Nginx -->|"Route: /api"
Proxy Pass| Backend

%% Backend connections
Backend -->|SQL Queries| DB
Backend -->|File Operations| Uploads

%% Styling
classDef cloudflare fill:#f96,stroke:#333,stroke-width:2px
classDef proxy fill:#6c9,stroke:#333,stroke-width:2px
classDef app fill:#69f,stroke:#333,stroke-width:2px
classDef data fill:#f9c,stroke:#333,stroke-width:2px
classDef user fill:#9cf,stroke:#333,stroke-width:2px

class CF,Edge cloudflare
class Socat,Nginx proxy
class Frontend,Backend app
class DB,Uploads data
class User,Internet user

```

Network Flow

```

sequenceDiagram
    participant U as 👤 User
    participant CF as ☁ Cloudflare
    participant S as 🔄 Socat
    participant N as ⚙ Nginx
    participant FE as 🌈 Frontend
    participant BE as < Backend
    participant DB as 🗄 Database

    Note over U,DB: Initial Page Load
    U->>CF: GET https://python-neon-votes-flu.trycloudflare.com/

```

```

CF-->>S: Forward to localhost:5173
S-->>N: Forward to localhost:3000
N-->>FE: Serve index.html + assets
FE-->>N: Static files
N-->>S: Response
S-->>CF: Response
CF-->>U: HTML + JS + CSS

Note over U,DB: Login Request
U-->>CF: POST /api/auth/login
CF-->>S: Forward request
S-->>N: Forward request
N-->>BE: Proxy to localhost:5001/api/auth/login
BE-->>DB: SELECT user credentials
DB-->>BE: User data
BE-->>N: JSON response + JWT token
N-->>S: Response
S-->>CF: Response
CF-->>U: Login successful

```

```

Note over U,DB: Document Upload
U-->>CF: POST /api/documents (multipart/form-data)
CF-->>S: Forward request
S-->>N: Forward request
N-->>BE: Proxy upload request
BE-->>DB: INSERT document metadata
BE-->>Uploads: Save file to disk
DB-->>BE: Success
Uploads-->>BE: File saved
BE-->>N: Success response
N-->>S: Response
S-->>CF: Response
CF-->>U: Upload complete

```

Port Mapping

```

graph LR
    subgraph "Port Configuration"
        P5173[Port 5173]
        Socat[Listener]
        P3000[Port 3000]
        Nginx
        P5001[Port 5001]
        Backend[API]
        P5432[Port 5432]
        PostgreSQL
    end

    P5173 -->|TCP Forward| P3000
    P3000 -->|HTTP Proxy /api| P5001
    P5001 -->|Database Connection| P5432

```

```
style P5173 fill:#ff9,stroke:#333,stroke-width:2px
style P3000 fill:#9f9,stroke:#333,stroke-width:2px
style P5001 fill:#99f,stroke:#333,stroke-width:2px
style P5432 fill:#f9f,stroke:#333,stroke-width:2px
```

Technology Stack

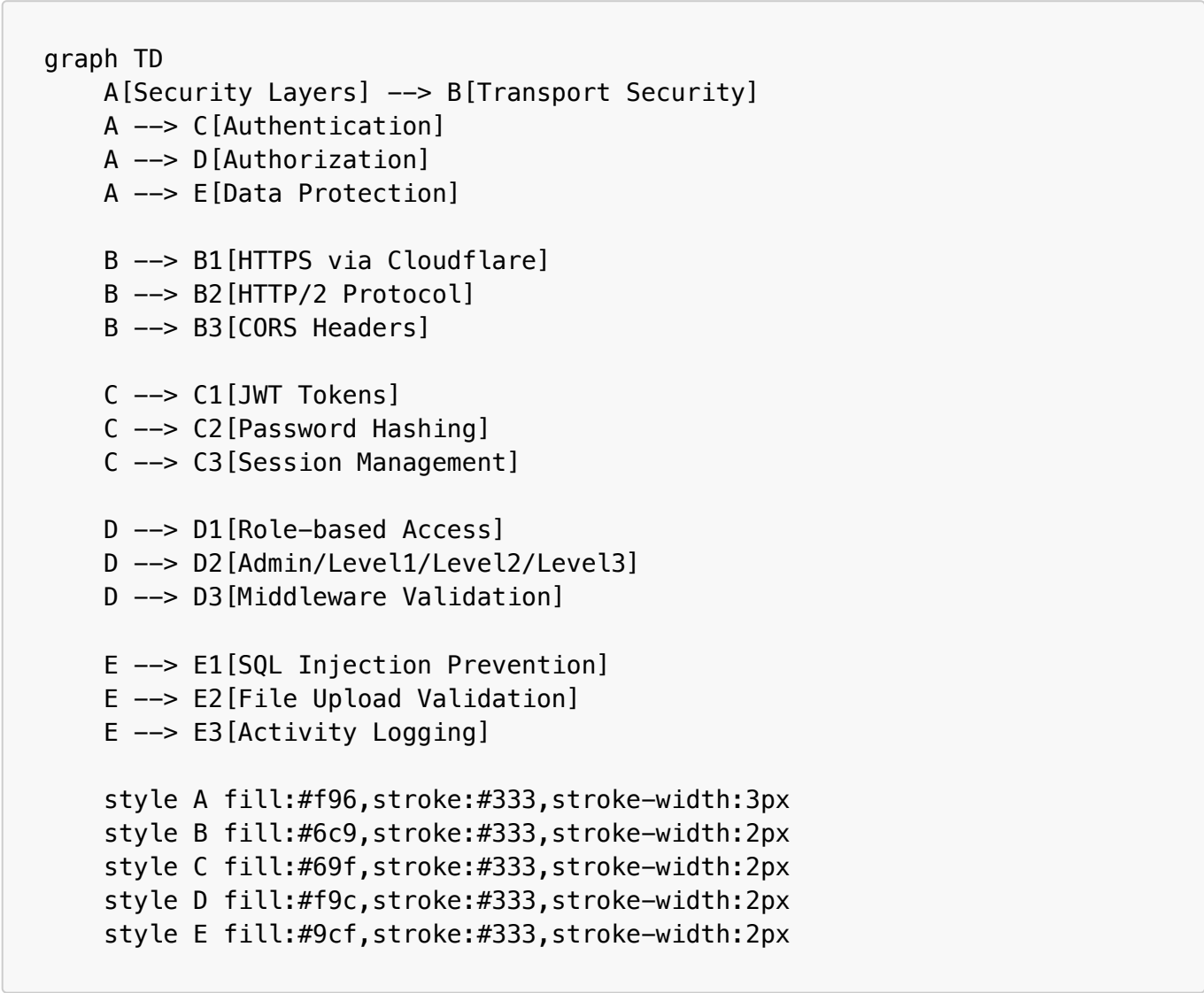
```
mindmap
  root((DocuMan
  System))
    Frontend
      React 18
      TypeScript
      React Router DOM
      Vite 7.2.2
    Backend
      Node.js 20.19.5
      Express.js
      Sequelize ORM
      JWT Auth
      Multer Upload
    Database
      PostgreSQL
      Models
        Users
        Documents
        SubDocuments
        ActivityLogs
    Infrastructure
      Nginx 1.29.3
      Socat TCP Proxy
      Cloudflare Tunnel
      macOS Darwin
    Security
      JWT Tokens
      Password Hashing
      CORS Headers
      Role-based Access
```

Deployment Architecture

Component	Technology	Port	Purpose
Public Access	Cloudflare Tunnel	HTTPS	Public internet gateway
Port Forwarder	Socat	5173	TCP port forwarding
Reverse Proxy	Nginx	3000	Route requests, serve static files

Component	Technology	Port	Purpose
Frontend	React + Vite Build	-	Single Page Application
Backend API	Node.js + Express	5001	REST API server
Database	PostgreSQL	5432	Data persistence
File Storage	Local Filesystem	-	Document storage

Security Features



Access Information

Public URL: <https://python-neon-votes-flu.trycloudflare.com>

Note: URL changes on every tunnel restart (Quick Tunnel limitation)

Default Credentials:

- Username: `admin`
- Password: `admin123`

Services Status:

-  Frontend: Running (Static files via Nginx)
-  Backend: Running (Node.js on port 5001)
-  Database: Connected (PostgreSQL)
-  Cloudflare Tunnel: Active
-  Nginx: Running (Port 3000)
-  Socat: Running (Port 5173 → 3000)

Generated: November 27, 2025 **System:** DocuMan - Document Management System **Status:** Production Ready 